

HUYNH QUOC TIEN

Full Stack JavaScript / TypeScript Developer / Software Engineer

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SUMMARY

Software Engineer with a solid foundation in computer science and hands-on experience developing high-performance, real-time web systems. Proficient in **NodeJS**, **NestJS**, **TypeScript**, **PostgreSQL**, **Redis**, and **Next.js/React**, with a proven ability to resolve concurrency challenges, optimize database operations, and build end-to-end applications. Strong advocate for clean architecture, automated testing/CI-CD, and containerized deployment with **Docker**.

EDUCATION

Sai Gon University

Engineer's Degree in Software Engineering

2021 - 2026

GPA: 2.89/4.0

EXPERIENCE

Freelance Full-Stack Developer

Dec 2025 – May 2026

Chip-equipped Driving Test Booking System (Client Project)

Ho Chi Minh City

- Engineered a full-stack booking system for chip-equipped driving test simulations using **NestJS**, **Next.js 15**, **Prisma**, and **PostgreSQL**.
- Eliminated double-booking and race conditions by implementing a real-time slot-locking mechanism with **Redis** and **Socket.io**, broadcasting slot status with sub-100ms latency.
- Designed automated cron workers using **@nestjs/schedule** to auto-release expired pending reservations (15-min TTL), improving vehicle utilization and reducing slot holding idle time.
- Built an intuitive time-slot grid UI with **Next.js (App Router)** and **Tailwind CSS**, enabling real-time slot state reflection and seamless multi-vehicle schedule tracking.
- Integrated automated **VietQR** dynamic payment payload generation and secured role-based access control (**RBAC**) via **JWT** and custom Guards.

Backend Developer Intern

May 2025 – Nov 2025

VNA International Technology JSC, Thu Duc City, Ho Chi Minh City

- Developed a labor safety reporting system, reducing manual reporting time by 30% using **NestJS**, **TypeORM**, and **PostgreSQL**.
- Architected modular RESTful APIs with DTO/Service patterns and secured endpoints using **JWT** and custom Guards.
- Automated file processing by integrating **Multer-S3** with **MinIO**, implementing an auto-cleanup service for outdated files.
- Engineered management modules for user-roles, report configuration, and enterprise workflows.
- Integrated **nodemailer** for automated email notifications, improving user communication efficiency.
- Documented API endpoints using **Swagger**, facilitating a 20% faster frontend integration.

PROJECTS

Swarm.io - Multiplayer .io Game

ReactJS, Node.js, Socket.io, Redis

June 2026 – Present

- Developed an endless real-time multiplayer .io game, achieving 60 FPS performance for 50+ concurrent players.
- Engineered server-side game state management using Node.js and Socket.io, enabling synchronized real-time movement and interactions.
- Implemented advanced collision detection mechanics, optimizing damage calculations.
- Optimized real-time communication by integrating Redis Pub/Sub, reducing message latency by 50% for active users.
- Built a responsive game interface using ReactJS for seamless gameplay controls.

Esports Tournament Management Platform

ReactJS, NestJS, PostgreSQL, WebSocket, Docker

Jul 2025 – Jan 2026

- Architected a comprehensive esports platform, supporting 100+ concurrent tournaments with multi-role access control.
- Automated bracket generation for Single and Double Elimination formats, reducing manual administrative setup time by 80%.
- Implemented real-time match synchronization using Redis Pub/Sub, ensuring sub-100ms latency for bracket updates.
- Optimized PostgreSQL performance for hierarchical tournament data by applying recursive CTEs and indexing strategies.
- Containerized the stack with Docker and deployed a CI/CD pipeline using GitHub Actions, reducing deployment time by 50%.

Multiplayer Tetris Game

ReactJS, Node.js, Socket.io, CI/CD

Sep 2025 – Nov 2025

- Developed a real-time multiplayer Tetris game, supporting 10+ simultaneous players per room with sub-50ms synchronization latency.
- Implemented server-side logic to synchronize piece sequences, ensuring 100% state consistency across all clients.
- Managed server-side game state with dynamically increasing difficulty levels.
- Built the client using **ReactJS** to handle gameplay logic and real-time state rendering.
- Applied **Clean Code** and **HOCs** to improve component reusability by 25%.
- Configured a **CI/CD** pipeline, reducing deployment time by 40% for both client and server.

Ecommerce Website

ReactJS, MSSQL, Socket.io

Oct 2024 – Dec 2024

- Built a full-featured e-commerce platform, supporting 500+ daily requests with 99.9% uptime.
- Designed a secure user authentication system with RBAC, ensuring 100% protection of user data access.
- Integrated MSSQL for efficient data management, achieving sub-200ms product search times.
- Optimized server performance by implementing asynchronous operations in ASP.NET, reducing average page load time by 30%.
- Developed an intuitive admin dashboard to streamline inventory management, reducing order processing time by 20%.

TECHNICAL SKILLS

Backend: NestJS, TypeScript, Node.js, RESTful API, Socket.io, JWT

Frontend: ReactJS, HTML5/CSS3, Ant Design

Database: PostgreSQL, MySQL, MSSQL

DevOps & Tools: Docker, CI/CD, Git, Swagger, Postman